BUILDING / CONSTRUCTION MATERIALS

Introduction

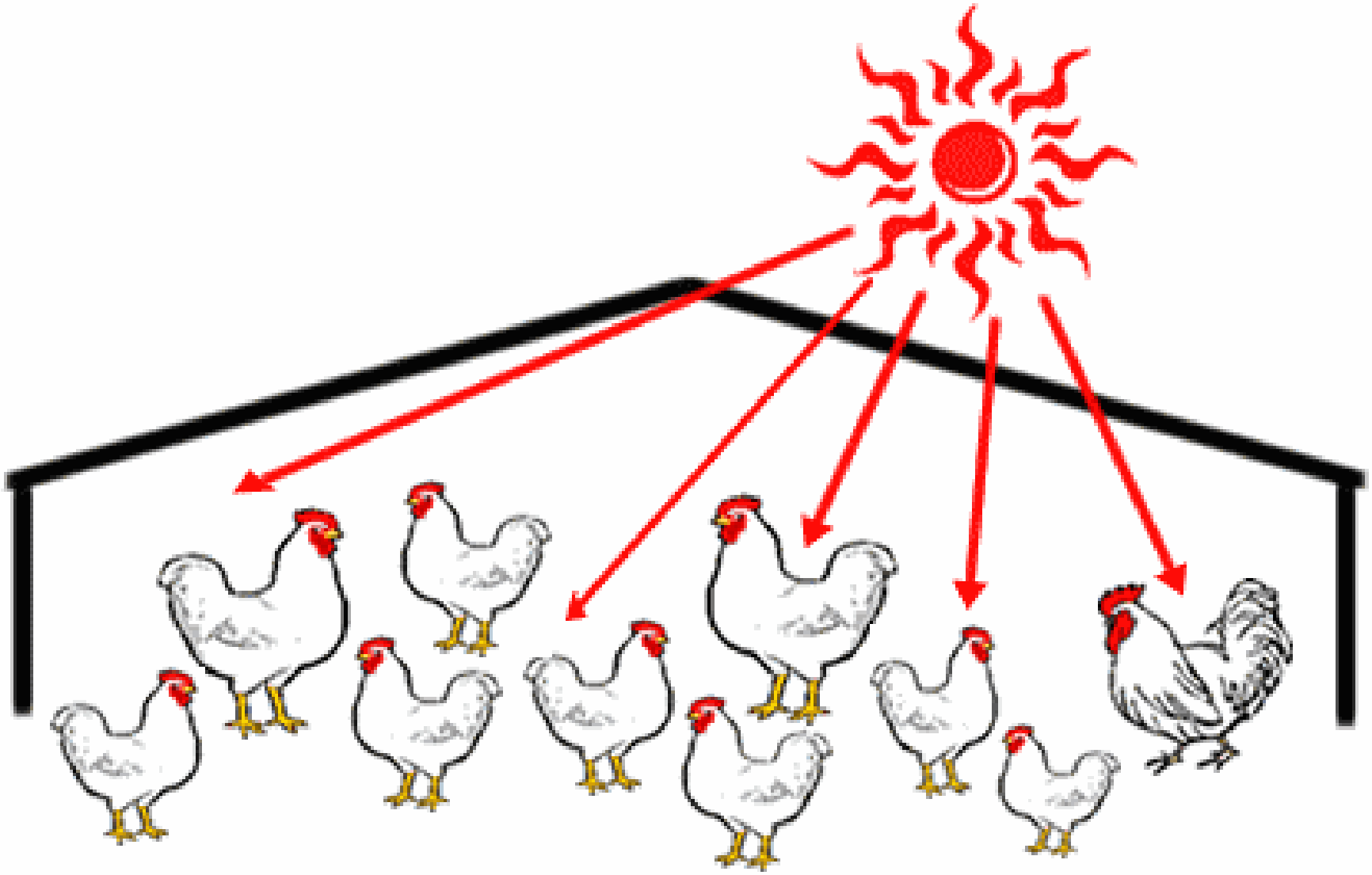
In the construction of Agricultural Structures, it is important to choose the right kind of construction material and the appropriate specification.

Factors that affect the choice of construction/building materials 1

- Type and function of structure
- Specific characteristics required of the material
- Economic aspects of the structure in terms of original investment and annual cost of maintenance
- Availability of material in the area or at the site
- Availability of local skilled labour to work with the desired material.

Factors that affect the choice of construction/building materials 2:

- Transportation cost and transportation possibilities
- Climatic conditions
- Selection of materials with properties dimensions and means of installation that are compatible
- Cultural acceptability or personal preference
- The suitable alternatives to replace material when necessary



Important Technical Properties of Construction Materials

- Density (bulk & specific)
- Porosity
- Strength (bending, compression, shear, tensile, modulus of Elasticity)
- Moisture or water diffusion property
- Wear resistance of materials
- Fire resistance property
- Sound insulation properties
- Moisture content
- Thermal properties (resistance, expansion, storage capacity)

WOOD (TIMBER)

- Timber is one of the oldest building materials along with stone, earth and various vegetable materials.
- It has remained until today the most versatile in terms of indoor comfort and health aspects.







The following reasons support the assertion that timber is the most versatile material:

- It is suitable for construction in all climatic zones.
- It is abundant and relatively cheap.
- It is easy to work with.
- It is compatible with traditional skills and does not necessarily require sophisticated equipment

- It is quite durable especially when treated.
- It is renewable i.e. when re-a forestation is done
- It provides good thermal insulation and sound absorption.
- Most timber species have reasonable high strength
- The production and processing of timber requires less energy than most other building materials.
- It has a nice appearance.

Certain characteristics of timber

- DENSITY

The density of timber ranges between 200kg/m^3 to 1250kg/m^3. This depends on specie and moisture content.

- NAIL HOLDING ABILITY

It has the ability to hold nail without much difficulty. This ability is greater for hard wood than for softer wood. Wood that is too hard is subject to splitting when nailed.

- PAINT HOLDING ABILITY

This differs between wood species and as a rule should be considered when selecting materials.

Certain characteristics of timber

- STRENGTH

Wood has a reasonably high strength to weight ratio.

- THERMAL PROPERTIES

It has an average heat insulation property

- SHRINKAGE

Shrinkage occurs if it is not well seasoned.

- RESISTANCE TO FIRE

Timber has a low fire resistance

Certain characteristics of timber

- RESISTANCE TO BIOLOGICAL DAMAGE

Biological damage/decay is mostly through insects and fungal attack

- RESISTANCE TO WIND

It is generally low but it can be improved when construction is properly done.

STRENGTH OF TIMBER

The strength of timber is affected by:

- Moisture content

Decreasing moisture content increases with strength

- The species

Different wood species have different strength properties

- Density
  - Strength increases with density
  - The number of strength reducing defects e.g.
- wide growth rings, knots.

DURABILITY OF WOOD

- Durability as applied to wood refers to the natural ability to resist the attacks of decaying organisms such as bacterial, fungi and insects.
- All sorts/species of wood from standing timber, logs of felled timber, stored and seasoned timber and the wood finally installed in a structure are subject to attack of many pathogens that cause incalculable damage to wood.

DURABILITY OF WOOD

- Relative durability of different wood species can be found through the “graveyard method”.
- In this method, test specimen 50 x 50 x 609mm is embedded in the ground leaving 228mm above the ground.
- Record of their life is taken.

Wood/timber preservation

- Preservation is used to protect timber from fungal decay, insect and fire attack.

This can be done in two ways:

1. Seasoning
2. Use of chemicals

Seasoning

To season a wood, the moisture content of the wood is reduced to its equilibrium moisture content. This safe moisture content is between 8% to 20% by weight and it depends on the timber species. The moisture content is however dependent on the timber species. Seasoning can be put in three categories:

- **Air seasoning/drying**
- **Forced air seasoning/drying**
- **Kiln drying**

Chemical treatment

Seasoning alone is not always sufficient to protect timber from fungal decay and insect attack. Protection from these biological hazards and fire is effectively achieved by preservative treatment with certain chemicals. Effective preservation depends on the preservative and how it is applied.

The following chemicals are used in the preservation of timber:

- Creosote
- Coal-tar
- Unleachable Metallic Salts
- Water Soluble Preservatives
- Organic Solvent Preservatives
- Emulsion Preservatives

Methods of preservation

- PRESSURE IMPREGNATION

The timber is put in a big container and the preservative is pumped through it under pressure. Preservatives 1, 3, and 4 are usually used.

- OPEN TANK TREATMENT

This is usually used for relatively small quantity of timber. The timber is soaked (steeped) in a tank with the preservative. It can be done hot or cold. Preservatives 1 and 3 are usually used.

Methods of preservation

- SUPERFICIAL PRESERVATION
  - Dipping, spraying and brushing applications come under superficial preservation.
  - They are not as effective as pressure impregnation and open tank treatment as the preservatives only penetrate the timber slightly. The wood must be seasoned and the surface should be dry and clean. Here we can use any of the preservative.
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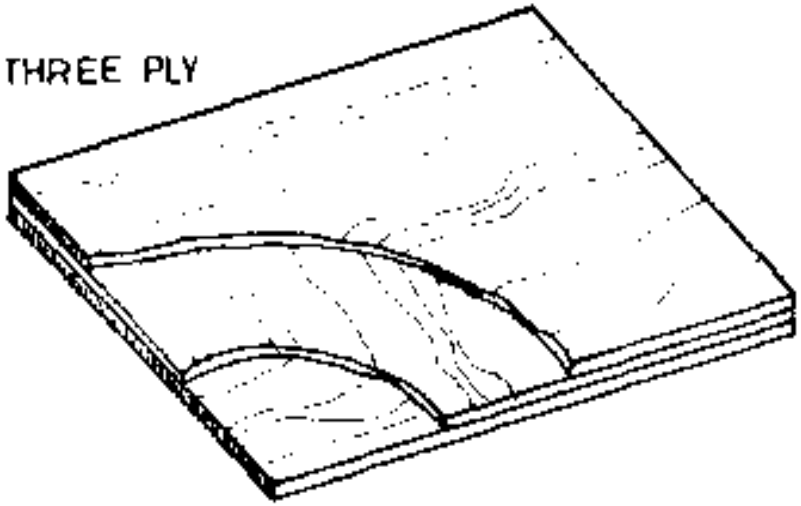
Products of timber

The three main products of timber are:

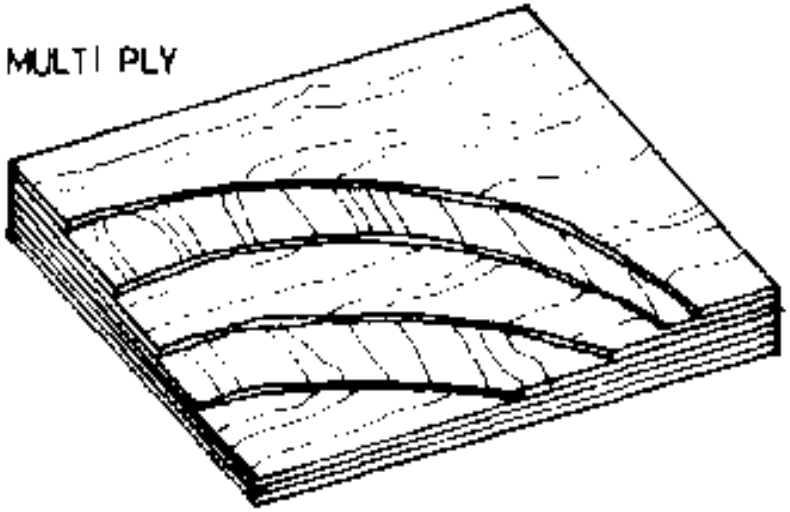
- Pole timber
- Sawn timber (Lumber)
- Timber manufactured production:

1. Plywood
2. Block boards
3. Particle boards/chip boards

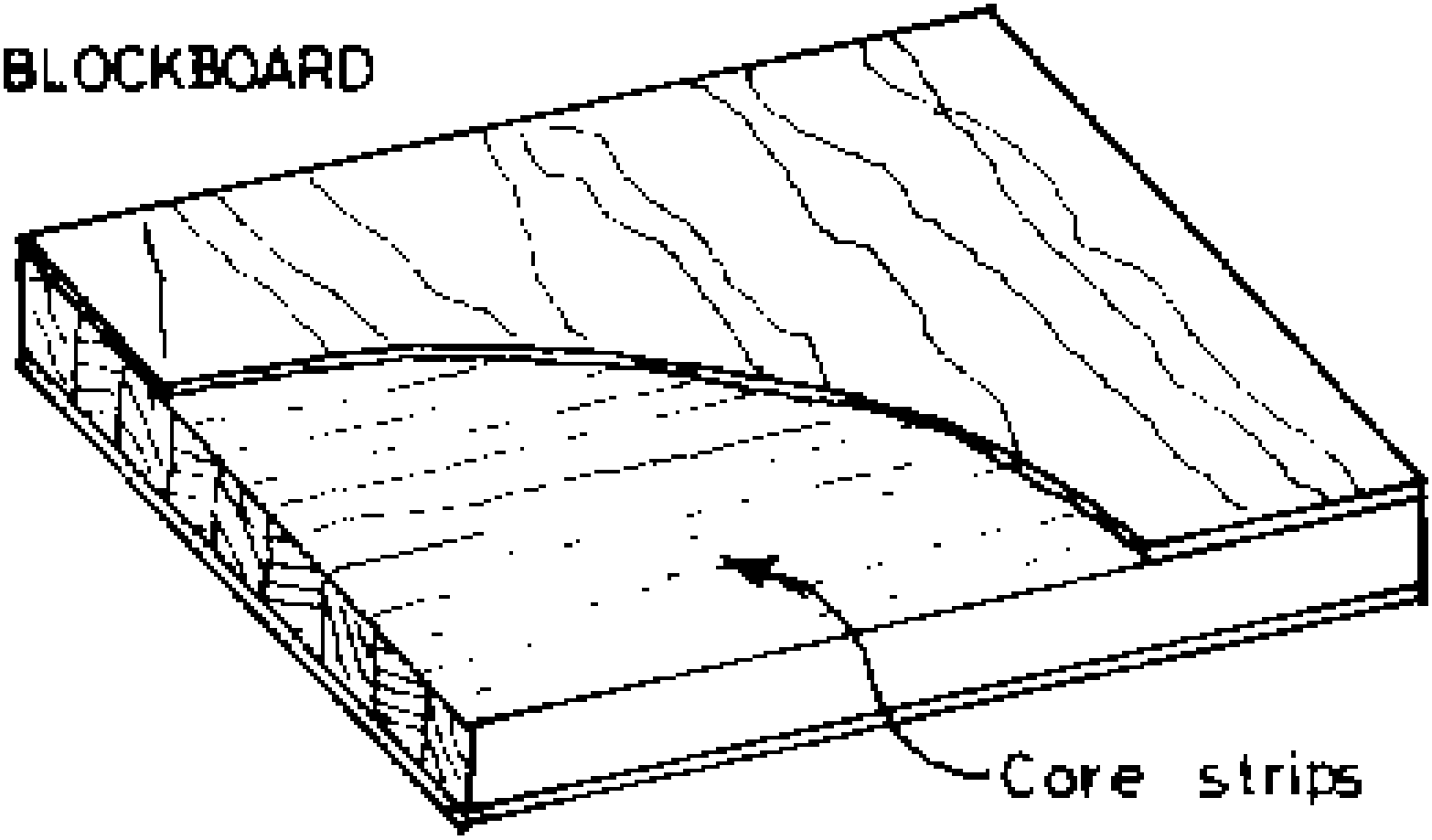
THREE PLY



MULTI PLY



BLOCKBOARD



Applications of timber

- Timber can be used for frame construction
- Timber can be use as beams and columns in construction
- It can be used as pole timber
- It can be used for walls and ceiling construction
- Timber can be used for furniture
- As construction material for agricultural structures, (silo, fences, houses and feeding trough, dryers, etc.)
- It can be used for doors, windows and stairs

Concrete

- Concrete is made by mixing cement paste, aggregates. The cement paste is made up of cement and water.




CONCRETE

Aggregate Classification

The aggregate used can range from fine to coarse. The aggregates that would pass through a sieve size of 5mm is referred to as fine aggregate. If it does not pass through the 5mm sieve, then it is referred to as coarse aggregate.

Fine aggregates include sand, crush gravel and crushed stones depending on how the crushing is done. Uncrushed gravels and small stones are referred to as coarse aggregates.

Points to note

- The quality of concrete is related to the binding quality of a paste.
- The most important property of fresh concrete is its' workability which relates to ease with which it can be placed and compacted.
- Wet concrete can take any shape.
- Concrete can have smooth or rough texture depending on the type of aggregate use.

Types of concrete

1. NORMAL CONCRETE

This is usually made up of mixture of fine aggregate, coarse aggregate water and cement. It is important to select and control the proportions of the materials used in other to have the correct strength and durability required. A nominal concrete mix of 1:2:4 should be used.

Types of concrete

1. **REINFORCED CONCRETE**

For reinforced concrete other dense aggregates are included in addition to the normal aggregate used in normal concrete. These may be steel, shot steel bar, iron rods and steel wire mesh.

1. **CONCRETE BLOCKS**

They are often made of 1:3:6 concrete mix with a maximum size aggregate of 10mm or a cement-sand mixture with a ratio of 1:7, 1:8 or 1:9.





Manufacture of concrete

- Concrete can either be made wholly on the site or factory made ready-mix or wholly secured pre-cast product.
- Production methods may vary from small, medium and large size sites.

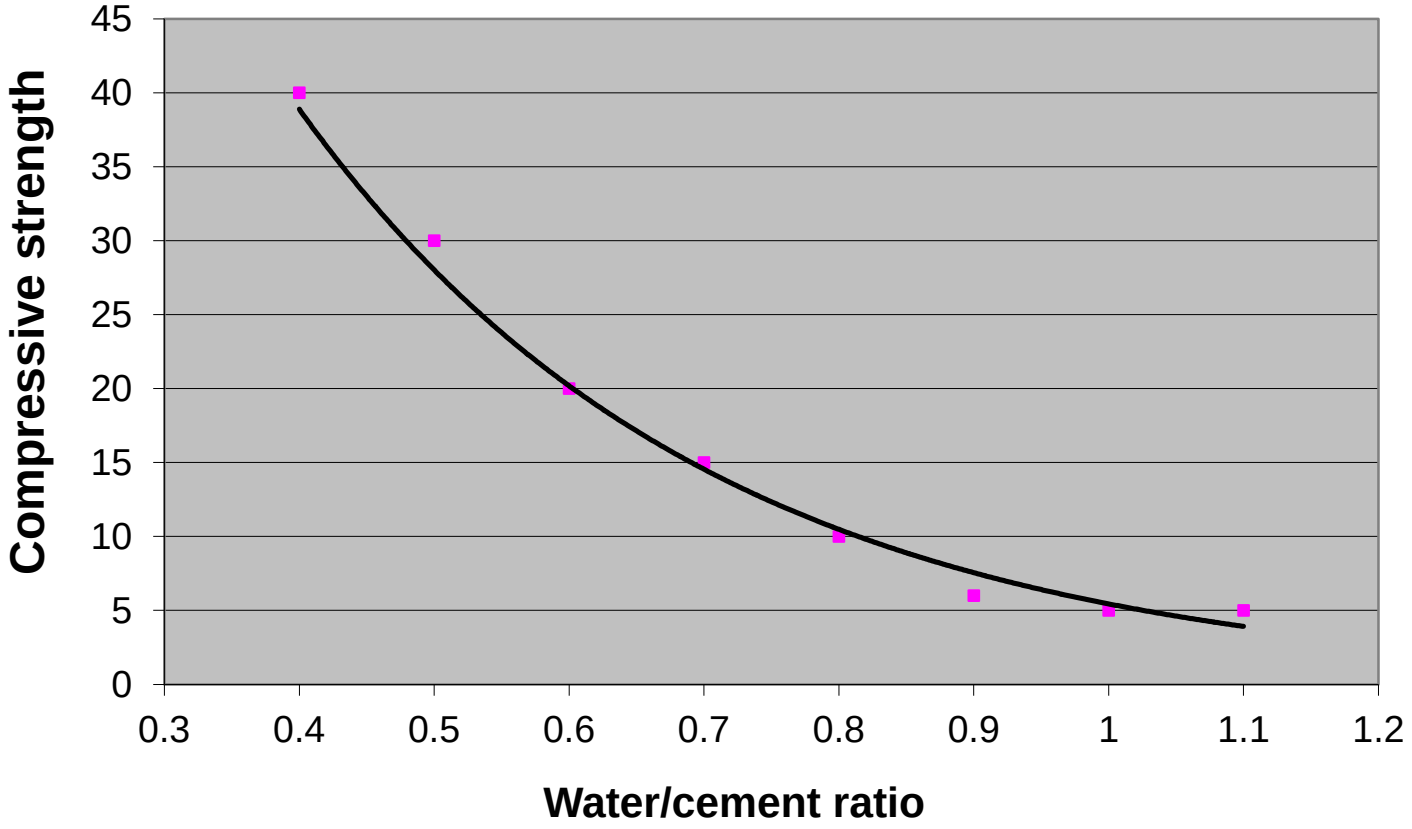
The processes of manufacture include:

- Storage of material
- Batching
- Mixing
- Formwork
- Compaction
- Curing
- Removal of formwork

Water: cement ratio

- The ratio of weight of water to the weight of cement required. The water: cement ratio determines the strength and durability of concrete.
- The amount of water contained in a batch is critical. A water cement ratio of 0.4 represents a proportion of 0.4 water to 1 cement.

The effect of water-cement ratio on the compressive strength is presented graphically below.



The strongest concrete is obtained with the lowest water-cement ratio which gives a workable mix that can be thoroughly compacted. Every 1% of water in excess of what is needed will reduce the strength by up to 5%. Water-cement ratio should however not be below 0.4:1 since this is the minimum required to hydrate the cement.

Cement: aggregate ratio

The aggregate: cement ratio is the ratio of the mass of aggregate to the mass of cement. This is used in conjunction with the water cement ratio for the determination of the concrete mixes.

A concrete mix of 1:2:4 represents the proportion of cement to fine aggregates to coarse aggregates.

Strength of concrete

Definition of strength

- The compressive strength of concrete is the characteristic strength of normal concrete allowed to season for 28 days.

- Load at failing and cross sectional area can be determined and hence the compressive strength.

Strength grading system

A grading system which corresponds to the characteristic crushing strength as used for the determination of concrete strength is shown below:

Grading System	N/mm^2
C7	$7.0N/mm^2$
C10	$10.0N/mm^2$
C20	$20.0N/mm^2$
C30	$30.0\ N/mm^2$
C40	$40.0\ N/mm^2$
C50	$50.0\ N/mm^2$
C60	$60.0\ N/mm^2$

The compressive strength of site concrete may differ from place to place because of:

- different curing (seasoning) condition
- different degree of compaction
- different methods of local application
- different dimensions of the aggregate particles

Advantages and Disadvantages

ADVANTAGES

- Concrete can take any shape
- Concrete has high compressive strength compared to other building materials
- Its high thermal capacities make it favourable for buildings in hot dry climates.
- Properly executed concrete is extremely durable, resistant to moisture penetration, fire and insect attacks.

Advantages and Disadvantages

DISADVANTAGES

- Cost of concrete is very high due to the use of cement.
- Corrosion of reinforced concrete is experienced in most coastal and moist climates
- Demolishing of concrete structure is very difficult
- It has the disadvantage of developing cracks.
- Algae growth.

Suggested use for various concrete grades and nominal mixes

Grade/Strength	Nominal mix	Use
C7 , 7N/mm2	1:3:8	Strip footing, trench fill
C10, 10N/mm2	1:4:6 1:3:6 1:4:5 1:3:5	Foundations, non-reinforced foundations, floors with very light traffic
C15, 15N/mm2	1:3:5	Foundation walls, basement walls
C20, 20N/mm2	1:3:4 1:2:4 1:3:3	Structural concrete, walls, reinforced floor slabs, Floors for dairy and beef cattle, pigs and poultry Septic tank, manure pits, water storage tanks, roads, driveways, pavings
C25, 25N/mm2	1:2:4	All concrete in milking parlours
C30, 30N/mm2	1:2:3	Dairies, silage silos, feed and drinking troughs
C35, 35N/mm2	1:1.5:3 1:1:2	Floors subjected to severe wear Roads and pavement for heavy machinery
C40, 40N/mm2		Concrete in very severe exposure
C50, 50N/mm2		Prefabricated structural elements

Mortar

Mortar is a plastic mixture of water and binding materials used to join concrete blocks, bricks or other masonry units It is desirable for mortar to hold moisture, be plastic enough to stick to the trowel and the blocks or bricks and finally to develop adequate strength without cracking.

Main Types:

- Lime mortar
- Cement mortar

Some metals used in construction

Steel



Uses

1. As metal sheets for wall construction
2. As roofing sheet (galvanized)
3. As Angle bars for construction of structures
4. As Steel beams for construction of structures
5. For manufacturing tool cutting edges
6. For the manufacturing of certain equipment and materials (bolts, screw, hinges locks, shovels, wheel barrows etc.)










Taiyu Industrial Group Co., Ltd

ALUMINIUM

Uses

1. For Roofing materials
2. For Aluminium Nails
3. For Storage tanks and pipes
4. For Roof gutter construction for rain harvesting




New Product Introduction
AlumaSteel®
Locking System

COPPER

Uses

1. For Plumbing
2. For Construction of hot H_20 and gas pipes
3. For manufacture of lightening conductors
4. Use in place of Fe or steel in places where corrosion is a problem.

Other local building materials



Bamboo

Uses

1. For building wall structure
2. As beams and scaffolding material
3. For roof frame construction (roof truss)
4. As roofing material in roof construction
5. Used in some cases for the construction of floors, bridges, etc.


Pavilion with extended porch



Other uses

- Furniture, mats,
- Fencing materials
- Pipes,
- Bamboo gutters,
- Silos (cribs) for crop storage
- Platforms for crop drying

Advantages

- 1.It is renewable
- 2.Has smooth surface hence there is no need of painting
- 3.Easy to handle
- 4.Cheap and abundant in growing areas
- 5.Use of simple manual methods with traditional tools in processing.
- 6.Provision of comfortable living conditions especially in hot climates.

Disadvantages

1.It is easily attacked by biological agents

2.It catches fire easily

3.Preservation is difficult with bamboo than timber

4.Bamboo causes greater tool wear than timber

Earth materials (mud)

Uses

1. House construction
2. Construction of various building elements
3. For foundation construction in areas of low water table
4. For floors construction (Dry, good drainage and areas with low water table)
5. Wall construction (examples):

- monolith walls
- masonry walls

6. Roof construction
7. For the manufacture of tiles
8. For flat roof construction







Earth materials (mud)

Advantages

1. Abundant and cheap
2. Easy to work with
3. Has relatively high fire resistance property
4. It is environmentally friendly
5. Requires low energy inputs in processing
6. It has a favourable climatic performance due to
7. its has low thermal conductivity
8. it has high thermal capacity
9. its low porosity

Earth materials (mud)

Disadvantages

- Wears very fast
- It shrinks and swells causing cracks and eventual collapse
- It has low resistance to impact
- Low resistance to wear
- Low acceptability among most social groups

Thatch





Thatch





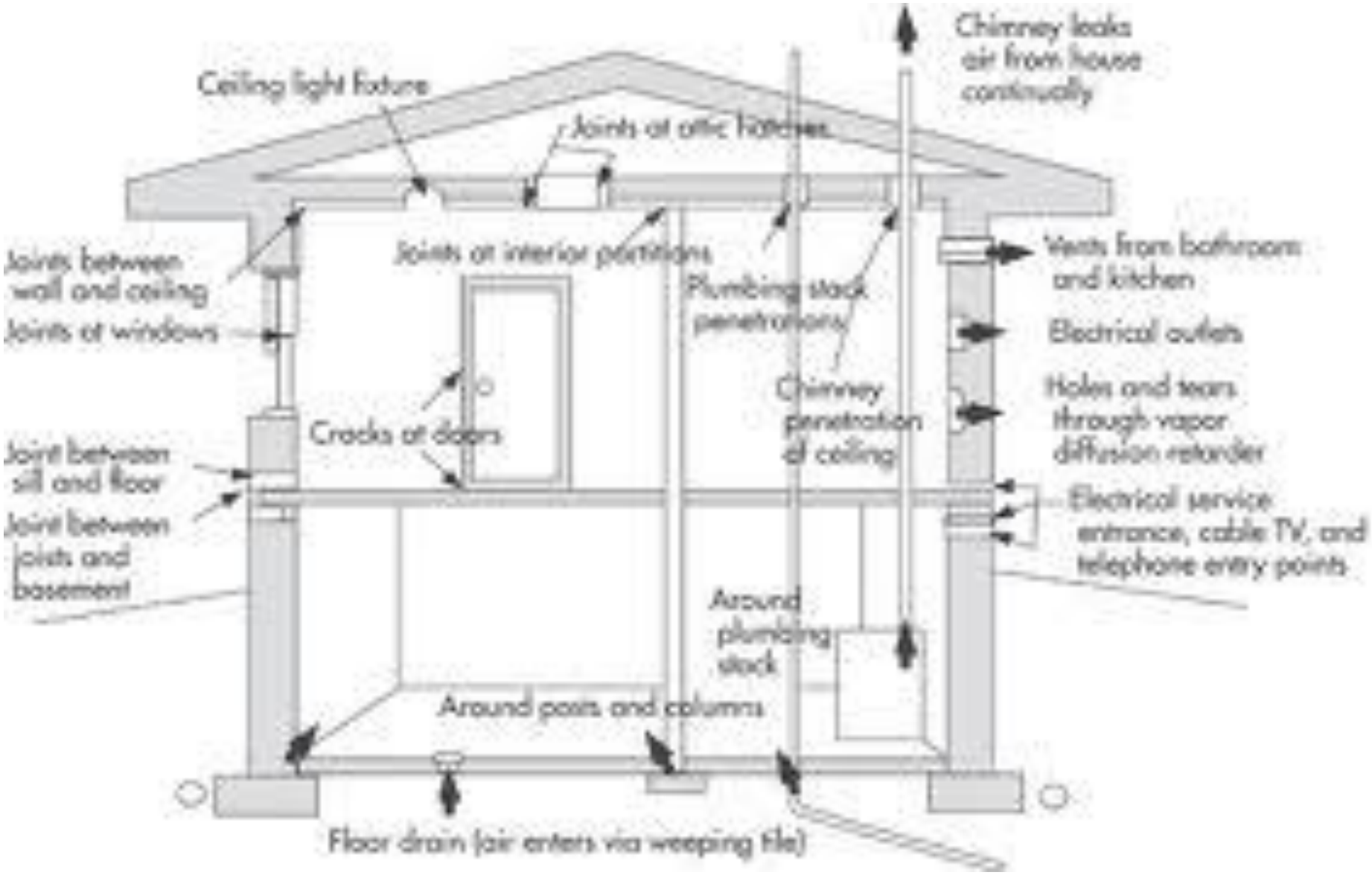




Building elements

The four major elements of a building:

1. Foundation
2. Floor
3. Wall
4. Roof


Chimney leaks air from house continually
Ceiling light fixture
Joints at attic hatches
Joints between wall and ceiling
Joints at windows
Joints at interior partitions
Plumbing stack penetrations
Vents from bathroom and kitchen
Electrical outlets
Holes and tears through vapor diffusion retarder
Chimney penetration at ceiling
Cracks at doors
Joint between sill and floor
Joint between joists and basement
Electrical service entrance, cable TV, and telephone entry points
Around plumbing stack
Around posts and columns
Floor drain (air enters via weeping tile)

Foundation

FUNCTIONS:

- Supports the building and the loads that are within or on the building.
- Serves as water or moisture barrier.

Types of foundations:

1. Pad foundation
2. strip foundation
3. slab foundation
4. pole foundation
5. step foundation

Floors

FUNCTIONS:

- Serves as moisture barrier
- Protection against rodents
- Prevention of termites and insects intrusion
- Support loads

Types of floors

1. Normal solid floors:
   - concrete floors
   - compacted soil floors
   - timber floors

Types of floors

2. Suspended floors : (Raised floors)

- Slatted floors;
  - wooden slabs, metal slabs, concrete slabs, mesh
- Slotted floors;
  - metal slots, concrete slot
- Concrete floors
  - Concrete
- Timber floor
  - timber

Materials

1. Earth materials (mud) 2.Concrete materials
2. Wood or bamboo 3.Metal sheet, metal slabs and wire mesh

Walls

FUNCTIONS:

- Security against human and animal intrusion
- Resistance to undesirable climatic conditions or weather (rain, wind, etc.)
- Privacy
- Support ceiling/roof structure
- Create comfort
- Provide thermal insulation.

Types of Walls

- Masonry wall (can be made of bricks/block)
- monolith wall (can be made of earth; concrete)
- Frame wall (can be made of stick, timber, bamboo, etc.)
- Cavity walls (can be made of bricks, blocks etc.)

Materials for wall construction

- Earth material
- Burnt brick
- Concrete block, Concrete
- Grass, bamboo, palm fronds
- Timber
- Metal sheets

Roofs

FUNCTIONS:

1. Used for protection against weather (rain, sun, dust, cold and heat.)
2. Provision of thermal insulation
3. Provision of fire protection

Types of roofs

1. FLAT ROOFS

Roofs with inclination of less than 10° are referred to as flat roofs.

Examples are:

- mud roof,
- concrete roofs .and
- roofs made from organic materials

Types of roofs

2. SLOPED ROOFS

Examples are:

- Single pitched roof
- Double pitched roof

3. CURVED ROOFS

For sloped roofs the following materials can be used;

- bamboo, wood, metals sheets, tiles and thatch.

Some building materials and their recommended angle of slope:

Material	Slope
Grass (thatch)	45°
Timber/bamboo	33°
Clay tiles	18° – 33°
Metal sheets	10° – 18°

Criteria for selection of roofing materials

The following criteria can be used for the selection of roofing materials:

1. Purpose of structure
2. Availability of the materials
3. Durability and cost
4. Technical properties
   - leak proof ability of the material
   - specific thermal characteristics of the material
   - resistance to fire property
5. Availability of people who can work with the material

Typical roof frame (ROOF TRUSS)

Wooden beams are usually used to construct the roof frame.

Bamboo, metal bars and rod can also be used.

Main components of gable roof frame

I. Rafters (Rf)

II. Purlins (PL)

III. Internal bracing (IB)

IV. The main tie or Lower cord (LC)

V. Ridge/ridge cap (R)

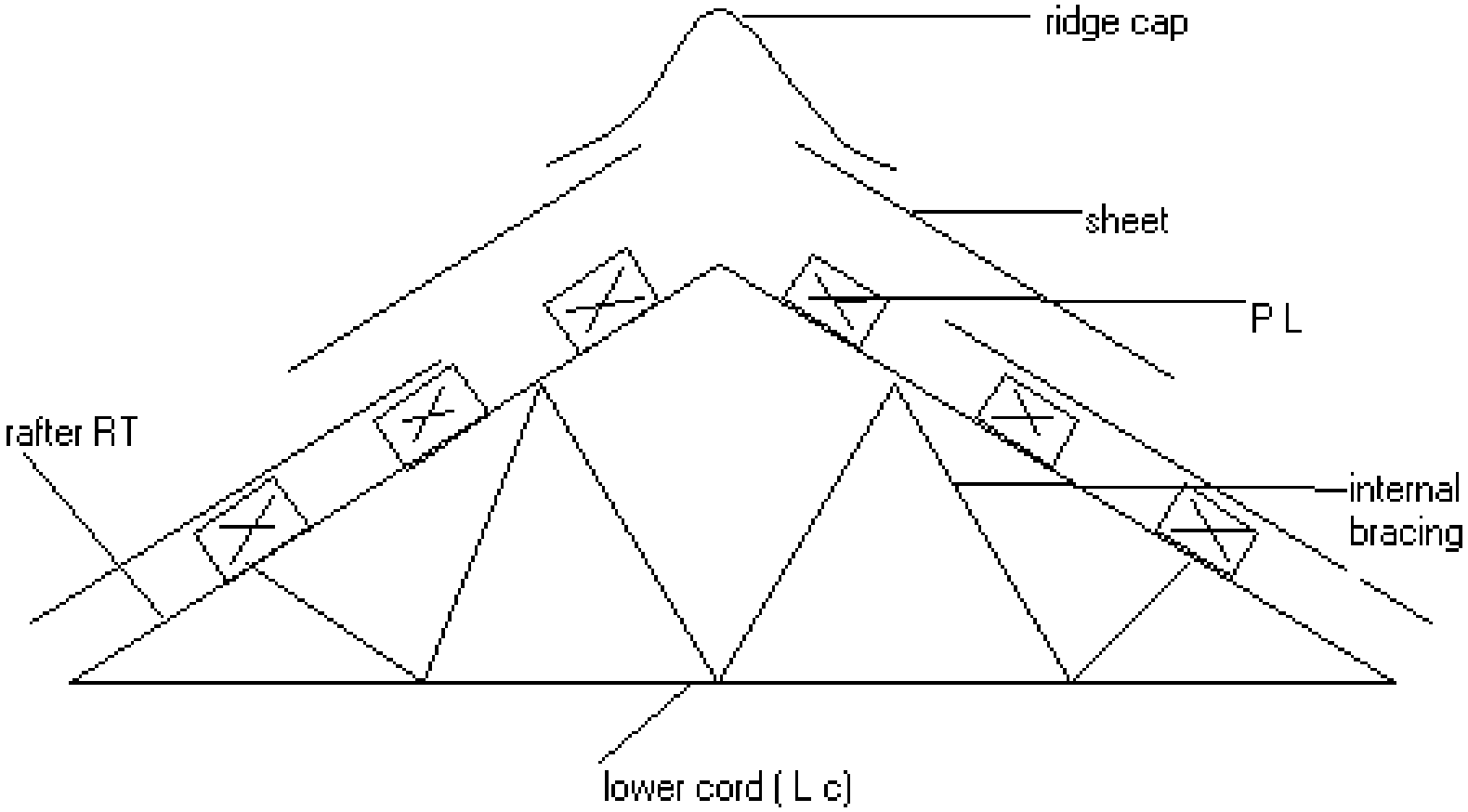


Figure 42: Components of gable roof

Deductions from a roof truss

1. Determination of load on all members can be achieved when the sketch is made.
    - Load on Purlin:
        - wind load, load due to sheets, self load of purlin
    - Load on Rafters:
        - wind load, load due to sheets, purlin load, self load of rafters

Design of the roof truss

1. Select the general layout of the truss members and truss spacing
2. Estimate external loads to be applied including the self weight of members, including wind loads

Design of the roof truss

3. Determine the critical or worse loading condition (dead load and other loads).
4. Analyse the framework to determine all the forces in all the members.
5. Select the materials and sections e.g. timber beam to produce in each member, a stress value which does not exceed the permissible value (ratio of the maximum bending moment to sectional modulus).